Asignment No :11

1) **Create a view that shows all of the customers who have the highest ratings.**

```
KD3_chandrakant_96233>create view mrating AS
    -> select * from customers where rating=(select MAX(rating) from customers);
Query OK, 0 rows affected (0.05 sec)

KD3_chandrakant_96233>select * from rating;
ERROR 1146 (42S02): Table 'sales.rating' doesn't exist
KD3_chandrakant_96233>select * from mrating;
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
2 rows in set (0.04 sec)
```

2) **Create a view that shows the number of salespeople in each city.**

```
KD3_chandrakant_96233>create view nosalepeople as
    -> select city  ,count(*) from salespeople
    -> group by city;
Query OK, 0 rows affected (0.04 sec)

KD3_chandrakant_96233>select * from nosalepeople;
+-----------+----------+
| city      | count(*) |
+-----------+----------+
| London    |        2 |
| San Jose  |        1 |
| Barcelona |        1 |
| New York  |        1 |
+-----------+----------+
4 rows in set (0.04 sec)
```

3) **Create a view that shows the average and total orders for each salesperson after his or her name. Assume all names are unique.**

```
KD3_chandrakant_96233>create view avgtot as
    -> select  s.sname,avg(o.amt),count(*)  from orders o INNER JOIN salespeople s on o.snu
m=s.snum
    ->  group by sname;
Query OK, 0 rows affected (0.05 sec)

KD3_chandrakant_96233>select * from avgtot;
+---------+-------------+----------+
| sname   | avg(o.amt)  | count(*) |
+---------+-------------+----------+
| Rifkin  |  558.425000 |        2 |
| Peel    | 5127.356667 |        3 |
| Motika  | 1900.100000 |        1 |
| Serres  | 1848.716667 |        3 |
| Axelrod | 1713.230000 |        1 |
+---------+-------------+----------+
5 rows in set (0.05 sec)
```

4) Create a view that shows each salesperson with multiple customers.

```
KD3_chandrakant_96233>CREATE VIEW salespeople_multiple_customers AS
    -> SELECT s.snum,
    ->        s.sname,
    ->        s.city,
    ->        COUNT(c.cnum) AS total_customers
    -> FROM salespeople s
    -> JOIN customers c
    -> ON s.snum = c.snum
    -> GROUP BY s.snum, s.sname, s.city
    -> HAVING COUNT(c.cnum) > 1;
Query OK, 0 rows affected (0.13 sec)

KD3_chandrakant_96233>SELECT * FROM salespeople_multiple_customers;
+------+--------+----------+-----------------+
| snum | sname  | city     | total_customers |
+------+--------+----------+-----------------+
| 1001 | Peel   | London   |               2 |
| 1002 | Serres | San Jose |               2 |
+------+--------+----------+-----------------+
```

5) Which of these views are updateable (will allow DML operations

ANSWER: Option 4 is correct because it has not group by , join or distinct like others option.

6) Create a view of the Salespeople table called Commissions. This view will include only the snum and comm fields. Through this view, someone could enter or change commissions, but only to values between .10 and .20

```
2 rows in set (0.02 sec)

KD3_chandrakant_96233>CREATE VIEW Commissions AS
    -> SELECT snum, comm
    -> FROM salespeople
    -> WHERE comm BETWEEN 0.10 AND 0.20
    -> WITH CHECK OPTION;
Query OK, 0 rows affected (0.06 sec)

KD3_chandrakant_96233>UPDATE Commissions
    -> SET comm = 0.18
    -> WHERE snum = 1001;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0

KD3_chandrakant_96233>select * from Commissions;
+------+------+
| snum | comm |
+------+------+
| 1001 | 0.18 |
| 1002 | 0.13 |
| 1004 | 0.11 |
| 1007 | 0.15 |
| 1003 | 0.10 |
+------+------+
5 rows in set (0.00 sec)

KD3_chandrakant_96233>select * from salespeople;
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1001 | Peel    | London    | 0.18 |
| 1002 | Serres  | San Jose  | 0.13 |
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
5 rows in set (0.00 sec)

KD3_chandrakant_96233>UPDATE Commissions
    -> SET comm = 0.18
    -> WHERE snum = 1001;
Query OK, 0 rows affected (0.04 sec)
Rows matched: 1  Changed: 0  Warnings: 0

KD3_chandrakant_96233>select * from Commissions;
+------+------+
| snum | comm |
+------+------+
| 1001 | 0.18 |
| 1002 | 0.13 |
| 1004 | 0.11 |
| 1007 | 0.15 |
| 1003 | 0.10 |
+------+------+
5 rows in set (0.00 sec)

KD3_chandrakant_96233>select * from salespeople;
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1001 | Peel    | London    | 0.18 |
| 1002 | Serres  | San Jose  | 0.13 |
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
```